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Review

Optimal extent of lateral neck dissection for well-differentiated thyroid carcinoma with metastatic lateral neck lymph nodes: A systematic review and meta-analysis

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ABSTRACT

The purpose of this systematic review and meta-analysis was to determine the optimal extent of lateral neck dissection in patients with well-differentiated thyroid carcinoma with clinically confirmed lateral neck lymph node metastases. All studies reporting the distribution of metastatic lymph nodes in level IIb or level V, complication rate, recurrence rate, or clinical outcomes according to the extent of lateral neck dissection were collected from MEDLINE and Embase databases. Two reviewers independently retrieved articles, extracted data, and assessed the quality of the studies. A total of 40 criteria-meeting studies were included in the systematic review and meta-analysis, representing a total of 6 027 patients. The distribution of metastatic lymph nodes was 13.7% (95% confidence interval [CI]: 8.2–21.9%) in level IIb and 22.1% (95% CI: 18.6–26.1%) in level V. Shoulder syndrome complication showed a tendency to increase when comprehensive neck dissection was performed. The recurrence rate was 11.2% (95% CI: 8.4–14.9%) in the comprehensive neck dissection group and 11.0% (95% CI: 4.2–26.1%) in the selective neck dissection group. Clinical outcomes showed no difference between groups. In conclusion, selective neck dissection may be considered in patients with well-differentiated thyroid carcinoma with lateral neck lymph node metastases without any other risk factors.

Introduction

Well-differentiated thyroid carcinoma (WDTC) is the most common malignancy in the head and neck area, and in 2014, more than 63,000 new cases were diagnosed in the United States. Similarly, in South Korea, more than 20,000 new cases of thyroid carcinoma are diagnosed each year, and the incidence is increasing. In WDTC patients, nodal metastases have been reported in 30–60% of cases [1–3], and have also been reported in cases of small and intrathyroidal primary tumors [4].

Several studies have reported that regional lymph node (LN) metastases generally do not show a clinically significant effect on survival in WDTC patients. On the other hand, more recent publications have reported that metastatic disease in the lateral compartment is associated with poorer outcomes, particularly in older patients [5]. Therefore, the effects of regional LN metastases on survival remains controversial and is closely related to other risk factors such as age, primary tumor size, LN yield, and extranodal extension [6–8]. However, it is well accepted that the degree of LN metastases in the lateral compartment is closely related to recurrence rate [9]. Therefore, lateral compartment LN metastases were included in the American Thyroid Association (ATA) guidelines in 2015 as an important factor in predicting structural recurrence [10].

In general, therapeutic lateral neck dissection is recommended for WDTC patients if lateral compartment LN metastases are confirmed by clinical examination or radiological evaluation [10]; this has not been a subject of debate. However, there are no definitive guidelines for the extent of lateral neck dissection of the confirmed metastatic side. Although comprehensive therapeutic neck dissection (level II to level V) has been generally accepted, comprehensive neck dissection in patients without extensive LN metastases may not be effective when considering the balance of oncologic safety and functional outcome. Two anatomical neck levels have been particularly controversial in therapeutic

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lateral neck dissection in WDTC patients: level IIb, which includes the upper jugular posterior to the spinal accessory nerve; and level V, which is located between the skull base and the clavicle behind the posterior edge of the sternocleidomastoid muscle. This is because level IIb and level V dissection are the main procedures leading to postoperative complications due to injuries of the spinal accessory nerve [11,12]. Therefore, a number of studies have examined the distribution of metastatic LN and the possibility of omitting levels IIb and V during neck dissection [13–19]. However, there have been no clear conclusions, and this remains a subject of ongoing debate.

The aim of this systematic review and meta-analysis was to present the appropriate neck dissection range for WDTC patients clinically diagnosed as lateral LN metastases. We analyzed the literature reporting on the distribution of LN metastases at level IIb and level V. In addition, we analyzed studies comparing complication rates, recurrence rates, and clinical outcomes between comprehensive neck dissection and selective neck dissection omitting the controversial sites.

Material and methods

Search strategy

Literature searches were performed using MEDLINE and Embase databases from 1966 to July 2018. Literature obtained by exploring the medical subject headings “thyroid tumor” or “papillary carcinoma” or “follicular carcinoma” was collected into the first group. The following search terms were used for the second group: “lymph node” or “lymphadenopathy” or “lymph node metastasis” or “neck dissection” or “lymph node dissection.” The medical subject headings were modified to include all related terms after checking the controlled vocabulary associated with the search terms in MeSH and EMTREE. Finally, the two groups were cross-referenced and limited to English-language, human adult studies. This process resulted in a total of 5 162 articles, which were reviewed by the following three stages.

Review process

The review was conducted in three stages, which were performed using two independent reviewers. The first stage was reviewing titles and the second stage was reviewing abstracts. Finally, for the third stage, full articles were obtained reviewed. All stages were performed on the basis of the following inclusion and exclusion criteria, and controversial papers were concluded by agreement between the two reviewers.

Inclusion criteria

- All studies on patients with WDTC with lateral neck disease undergoing therapeutic neck dissection and/or central LNs.
- All studies reporting at least one of the following related data types: LN distribution, clinical outcome, recurrence rate, or complications.
- Studies reporting lateral neck dissection by the open method.

Exclusion criteria

- Review articles, commentaries, technique articles, and case reports with a single patient.
- Studies in which the method of neck dissection or its range is unclear.
- Secondary operations or salvage operations in patients with recurrent disease.

Data extraction

Data extraction focused on the aims of this study. Extracted data included: percentage of LN distribution with involvement at levels II b and V; study design; sample size; evaluation methods to identify metastatic LN; range of lateral neck dissection; follow-up time; adjuvant treatment; overall survival (OS), disease-specific survival (DSS), disease-free survival (DFS), recurrence rate, and complication rate; and p-value. All data were collected by two independent reviewers and synthesized.

Risk of bias analysis

The risk for bias of selected studies was assessed using the Risk of Bias Assessment Tool for Non-randomized Studies (RoBANS) for quality assessment of non-randomized experimental studies. This tool evaluates a total of six items and includes the selection of participants, confounding variables, measurement of intervention, blinding for outcome assessment, incomplete outcome data, and selective outcome reporting. The evaluation for each item is measured as low risk, high risk, or unclear risk. Two reviewers evaluated each item independently, and any unmatched results were re-evaluated. The results of this assessment were summarized using Review Manager 5 (Cochrane, London, UK).

Quantitative data analysis

LN distribution was measured using extracted data corresponding to level IIb and level V. The confidence interval (CI) was calculated using the Comprehensive Meta-Analysis software version 3 (Biostat, Inc., Englewood, NJ, USA), which utilizes distribution, using the event rate at each level. In the same way, data on each recurrence rate according to LN dissection range were collected and quantitative analyses were performed. Findings are presented as percentages. Data on clinical outcomes and complications related to LN dissection range were collected, but quantitative analyses were not performed because of the limited number of studies for the meta-analysis.

Results

Study selection

Throughout the three-stage process, 40 of 5162 studies matched the inclusion criteria and were selected and analyzed. Data corresponding to level IIb metastatic LN distribution were extracted from 11 studies [13–15,17,19–25], and data corresponding to level V metastatic LN distribution were extracted from 28 studies [6,13–39]. Because only three studies divided the metastatic LN metastasis distribution of level V into levels Va and Vb, the LN distribution of the whole level V was analyzed. Complication rates after LN dissection were analyzed in 11 studies [6,26,27,40–47]. Recurrence rate was extracted from a total of 19 studies [6–8,26–31,33,40–46,48,49], and clinical outcomes were extracted from 5 studies [28–30,42,49]. Individual studies were used for multiple outcome parameters. Fig. 1 shows the exclusion criteria and the number of studies excluded during each stage of review.

Metastatic LN distribution rate

A total of 11 studies were included in the meta-analysis of level IIb metastatic LN distribution, including three prospective studies. We included only those studies that confirmed LN metastasis in the lateral compartment via pre-operative ultrasound, fine needle aspiration biopsy, or computed tomography (Supplementary Table 1). A random-effects meta-analysis was performed, and the results are shown in Table 1. Level IIb was involved with metastasis in 13.7% (95% CI: 8.2–21.9%) of cases. The analyses of level V metastatic LN distribution included 28 studies, which are summarized in Supplementary Table 2. The random-effects meta-analysis showed that level V was involved with metastasis in 22.1% (95% CI: 18.6–26.1%) of cases (Table 2). Three studies reported the distribution of LN metastases from level Va, with a distribution ranging from 0% to 13% [17,22,24].

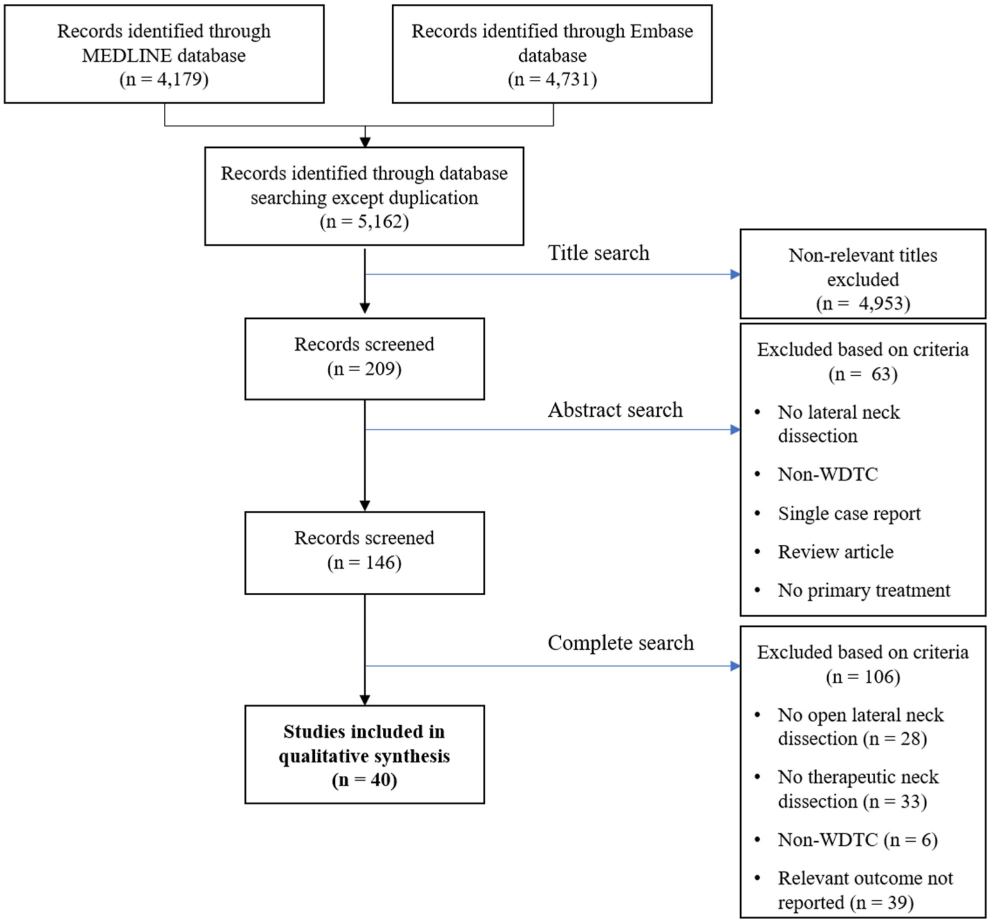


Fig 1. Flow diagram showing the selection of studies for inclusion in the systematic review and meta-analysis.

Complication rates according to the extent of lateral neck dissection

A total of 11 studies reported complication rates after neck dissection. Data were extracted from six types of complications: permanent recurrent laryngeal nerve palsy, permanent hypocalcemia, hematoma, shoulder syndrome, Horner syndrome, and chyle leak (Table 3). The incidence of shoulder syndrome was reported in seven studies, ranging from 1.5% to 27% for comprehensive neck dissection and from 0% to 2.7% for selective neck dissection [6,26,27,40,42,45,47]. Two studies showed a notably higher incidence of shoulder syndrome in the comprehensive neck dissection group [26,42].

Table 1
Percent of lymph nodes with involvement at level IIb.

Study name (year)	Statistics for each study				
	Event rate	Lower limit	Upper limit	Z-value	*p*-value
Kim et al. (2016)	0.104	0.075	0.142	-11.888	0.000
Lim et al. (2012)	0.100	0.053	0.181	-6.253	0.000
King et al. (2011)	0.615	0.456	0.753	1.423	0.155
Vayisoglu et al. (2010)	0.021	0.003	0.136	-3.777	0.000
Farrag et al. (2009)	0.085	0.036	0.188	-5.090	0.000
Koo et al. (2009)	0.118	0.062	0.212	-5.657	0.000
Roh et al. (2008)	0.167	0.089	0.291	-4.405	0.000
Lee et al. (2008)	0.068	0.040	0.114	-9.108	0.000
Yanir and Doweck (2008)	0.071	0.018	0.245	-3.495	0.000
Lee et al. (2007)	0.218	0.128	0.346	-3.911	0.000
Pingpank et al. (2002)	0.206	0.102	0.373	-3.182	0.001
Random	**0.137**	**0.082**	**0.219**	**-6.306**	**0.000**
I^2=86.042%					

Event rate and 95% CI

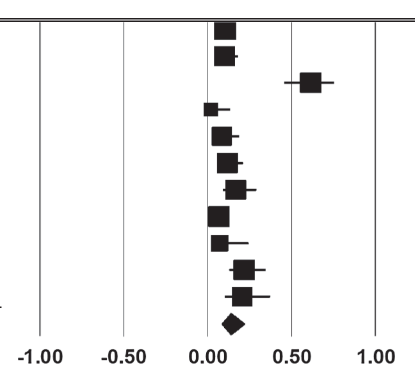
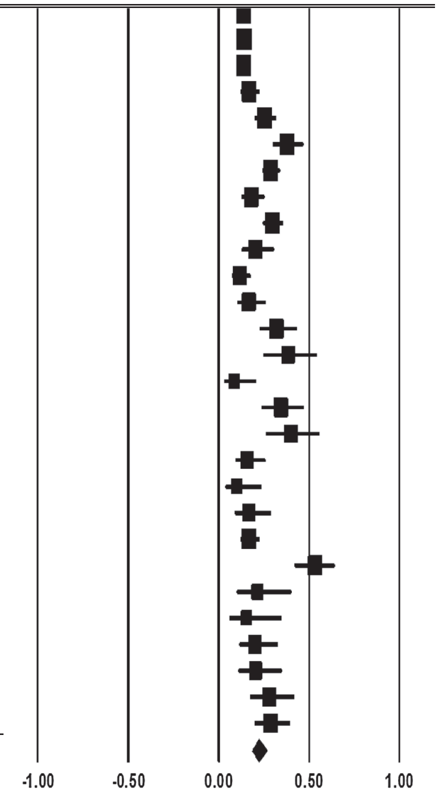


Table 2
Percent of lymph nodes with involvement at level V.

Study name (year)	Statistics for each study					Event rate and 95% CI
	Event rate	Lower limit	Upper limit	Z-value	*p*-value	
Kim et al. (2017)	0.139	0.114	0.168	-16.035	0.000	
Kim et al. (2017)	0.141	0.116	0.170	-16.132	0.000	
Kim et al. (2016)	0.138	0.105	0.180	-11.426	0.000	
Javid et al. (2016)	0.169	0.123	0.228	-8.335	0.000	
kang et al. (2014)	0.254	0.199	0.317	-6.790	0.000	
O'Neill et al. (2014)	0.380	0.298	0.470	-2.612	0.009	
Zhang et al. (2013)	0.288	0.242	0.339	-7.446	0.000	
Shim et al. (2013)	0.182	0.127	0.254	-6.934	0.000	
Merdad et al. (2012)	0.297	0.243	0.357	-6.200	0.000	
Keum et al. (2012)	0.203	0.128	0.306	-4.890	0.000	
Park et al. (2012)	0.116	0.073	0.179	-7.885	0.000	
Lim et al. (2012)	0.167	0.103	0.259	-5.686	0.000	
Wu et al. (2012)	0.321	0.227	0.431	-3.097	0.002	
King et al. (2011)	0.385	0.247	0.544	-1.423	0.155	
Vayisoglu et al. (2010)	0.085	0.032	0.206	-4.543	0.000	
Yüce et al. (2010)	0.344	0.236	0.471	-2.395	0.017	
Farrag et al. (2009)	0.400	0.262	0.557	-1.256	0.209	
Koo et al. (2009)	0.158	0.092	0.258	-5.320	0.000	
Iqbal et al. (2009)	0.100	0.038	0.238	-4.169	0.000	
Roh et al. (2008)	0.167	0.089	0.291	-4.405	0.000	
Lee et al. (2008)	0.168	0.121	0.227	-8.275	0.000	
Kupferman et al. (2008)	0.533	0.420	0.642	0.571	0.568	
Yanir and Doweck (2008)	0.214	0.099	0.402	-2.823	0.005	
Ahn et al. (2008)	0.154	0.059	0.346	-3.135	0.002	
Lee et al. (2007)	0.200	0.114	0.326	-4.112	0.000	
Kupferman et al. (2004)	0.205	0.110	0.349	-3.629	0.000	
Pingpank et al. (2002)	0.280	0.173	0.419	-2.999	0.000	
Sivanandan et al. (2001)	0.288	0.102	0.373	-3.182	0.000	
Random	**0.221**	**0.186**	**0.261**	**-11.364**	**0.000**	
I^2=85.475%						



Recurrence rates according to the extent of lateral neck dissection

Nineteen studies reported recurrence rates after neck dissection, and are summarized in Supplementary Table 3. The meta-analysis was performed using the recurrence rate reported in 12 studies, in which comprehensive neck dissection was performed [6,8,26–31,33,43,48,49]. The analysis included only studies that performed comprehensive neck dissection from level II to level V. The results of the random-effects meta-analysis indicated a recurrence rate of 11.2% (95% CI: 8.4–14.9%) (Table 4). When lateral neck metastasis was preoperatively confirmed, six studies were performed with selective neck dissection except for one or more levels from level II to level V, and the same meta-analysis was performed [40–42,44,45,49]. The results of the random-effects model indicated a recurrence rate of 11.0% (95% CI: 4.2–26.1%) (Table 4).

Clinical outcomes according to the extent of lateral neck dissection

Five studies reported OS, DFS, or DSS as clinical outcomes (Table 5). Three of these studies reported clinical outcomes after comprehensive neck dissection from level II to level V [28–30]. OS was 91.8% and 95.8% in two studies [28,30], DSS was 98.5% in one study [28], and DFS was 67.7% and 66% in two studies [28,29]. One study compared the clinical outcomes of radical neck dissection and selective neck dissection from level II to level IV, and there were no statistical differences between OS and DFS [49]. In another comparative study of comprehensive neck dissection and selective neck dissection, the overall recurrence-free probabilities were 94.4% and 89.4%, which was not statistically significant, although there was a significant difference in dissected neck field recurrence-free probability according to the extent of neck dissection (97.7% vs. 89.4%) [42].

Assessment of study quality and bias

The results of the evaluation of the risk for bias in the 40 studies included in this systematic review using the RoBANS tool are summarized in Fig. 2. In the 40 studies included in the assessment, 17.5% were evaluated as low risk in the selection of participants, and 22.5% were low risk for confounding variables. The low risk ratios of attrition and reporting biases were 15% and 20%, respectively. On the other hand, the high risk ratios of performance and detection biases were estimated to be 0% and 7.5%, respectively.

Study heterogeneity

Four meta-analysis results showed heterogeneity with I2 statistics from 84.944% to 86.042%. In other words, there was high heterogeneity among studies.

Discussion

The effects of locoregional LN metastasis on survival in WDTC patients remain controversial. According to studies in the SEER database, LN metastases, distant metastases, and large tumor sizes predicted lower OS in patients over the age of 45 years [50]. In addition, previous studies have reported that LN metastases in the lateral compartment,

Table 3
Complication rate according to lateral neck dissection range.

Author (Reference number)	Year	Study design	Extent of LN dissection (n)		Complication rate							Statistical comparison
					Permanent RLN palsy	Permanent hypocalcemia	Hema toma	Shoulder syndrome (SAN palsy)		Horner syndrome	Chyle leak	
								Transient	Permanent			
Zhang et al. [40]	2017	Prospective Randomized Study	NA	Level II, III, IV, Vb ND (32)	1/32 (3.1%)	0/32 (0%)	0/32 (0%)	1/32 (3.1%)		0/32 (0%)	0/32 (0%)	2/32 (6.3%)
Zhang et al. [41]	2017	Retrospective cohort study	NA	Level II, III, IV, Vb ND (38)	0/38 (0%)	0/38 (0%)	1/38 (2.6%)	NR		0/38 (0%)	2/38 (5.3%)	NR
Kim et al. [26]	2017	Retrospective cohort study	Level II-V ND (263)	Level II, III, IV ND (263)	NR	NR	0/263:1/263 (0%:0.4%)	24/263:7/263 (9.1%:2.7%) (persistent after 6 months from the surgery)		NR	5/263: 1/263 (1.9%:0.4%)	SAN palsy: $P = 0.002$
Kim et al. [6]	2017	Retrospective cohort study	Level II-V ND (658)	NA	0/658 (0%)	47/658 (7.1%)	1/658 (0.2%)	56/658 (8.5%) (persistent after 6 months from the surgery)		NR	11/658 (1.7%)	NR
McNamara et al. [42]	2016	Retrospective cohort study	Level II-Vb ± IIb, ND (364)	SND: III-IV ± II (120)	NR	NR	5/364: 2/120 (1.4%:1.7%)	22/364: 0/120 (6.0%:0%)	1/364: 0/120(0.3%: 0%)	NR	5/364: 2/120 (1.4%:1.7%)	SAN palsy: $P = 0.019$
Javid et al. [27]	2016	Retrospective analysis	Level II-V ND (195)	SND: II-IV (4) III-IV (2) IV (1)	0/195: -/- (0%:-)	NR	NR	3/195: -/- (1.5%:-)	0/195: -/- (0%:-)	0.195: -/- (0%:-)	2/195: -/- (1.0%:-)	NR
Lee et al. [43]	2015	Retrospective analysis	Level II-V ND (136)	NA	15/136 (11.0%)	8/136 (5.9%)	1/136 (0.7%)	NR		NR	4/136 (2.9%)	NR
Conzo et al. [44]	2013	Retrospective analysis	NA	Level III, IV, V ND (69)	1/69 (1.4%)	1/69 (1.4%)	NR	NR		NR	NR	NR
Yu et al. [45]	2012	Prospective Study	NA	Level II, III, IV (47)	NR	NR	1/47 (2.1%)	1/47 (2.1%)	0/47 (0%)	NR	2/47 (4.2%)	NR
Kang et al. [46]	2012	Retrospective analysis	Level IIa-Vb ± IIb,Va ND (109)	NA	0/109 (0%)	0/109 (0%)	3/109 (2.7%)	NR		0/109 (0%)	5/109 (4.6%)	NR
Kupferman et al. [47]	2004	Retrospective analysis	Level II-V ND (44)	NA	2/44 (4.5%)	0/44 (0%)	NR	12/44 (27.2%)	0/44 (0%)	NR	2/44 (4.5%)	NR

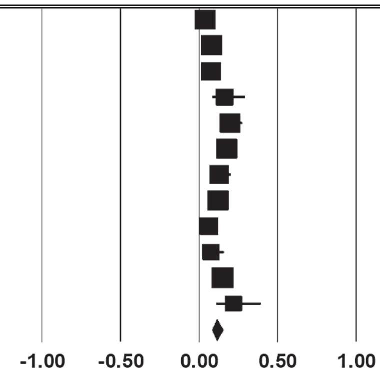
RLN, recurrent laryngeal nerve; SAN, spinal accessory nerve; LN, lymph node; ND, neck dissection; NA, not applicable; NR, not reported; SND, selective neck dissection.

Table 4
Recurrence rate according to lateral neck dissection range.

Comprehensive neck dissection
(Level II-V ND ± level I)

Study name (year)	Event rate	Lower limit	Upper limit	Z-value	*p*-value
Kim et al. (2017)	0.040	0.028	0.058	-15.841	0.000
Kim et al. (2017)	0.079	0.061	0.102	-16.994	0.000
Javid et al. (2016)	0.077	0.047	0.124	-9.246	0.000
Joo et al. (2015)	0.163	0.084	0.294	-4.230	0.000
Lee et al. (2015)	0.199	0.140	0.274	-6.492	0.000
Kang et al. (2014)	0.177	0.131	0.235	-8.480	0.000
O'Neill et al. (2014)	0.129	0.079	0.203	-6.892	0.000
Zhang et al. (2013)	0.121	0.090	0.161	-11.748	0.000
Shim et al. (2013)	0.063	0.033	0.116	-7.842	0.000
Keum et al. (2012)	0.076	0.034	0.159	-5.883	0.000
Ito et al. (2012)	0.150	0.126	0.178	-16.894	0.000
Turanli et al. (2007)	0.219	0.108	0.393	-2.975	0.000
Random	**0.112**	**0.084**	**0.149**	**-12.509**	**0.000**
I^2=85.887%					

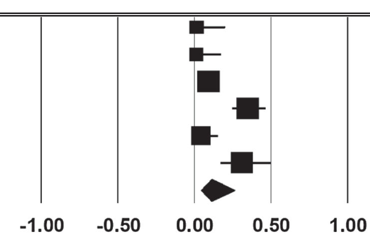
Event rate and 95% CI



Selective neck dissection
(One or more neck levels are excluded from ND)

Study name (year)	Event rate	Lower limit	Upper limit	Z-value	*p*-value
Zhang et al. (2017)	0.015	0.001	0.201	-2.929	0.003
Zhang et al. (2017)	0.013	0.001	0.175	-3.052	0.002
McNamara et al. (2016)	0.092	0.052	0.158	-7.249	0.000
Conzo et al. (2013)	0.348	0.245	0.467	-2.484	0.013
Yu et al. (2012)	0.043	0.011	0.155	-4.309	0.000
Turanli et al. (2007)	0.310	0.170	0.497	-1.993	0.046
Random	**0.110**	**0.042**	**0.261**	**-3.906**	**0.000**
I^2=84.944%					

Event rate and 95% CI



particularly in older patients, have a poorer clinical outcome than metastases in the central compartment [5,50–53]. However, the effects of LN metastasis on poor clinical prognosis are less than 2% in young patients and only 5% to 10% in older patients [53,54]. Furthermore, several studies have demonstrated that the mortality risk associated with LN metastasis is minimal in patients without distant metastases or gross extrathyroidal extension to prevertebral fascia or the carotid artery [5,55–57]. Therefore, LN metastasis of the lateral compartment was excluded from factors affecting N1b cancer staging in patients over 55 years old in the American Joint Committee on Cancer 8th edition (2018), unlike the 7th edition. Traditionally, LN metastasis in the lateral neck compartment is thought to have no significant effect on survival, but it affects the recurrence rate and is thought to be an important factor in recurrence in the ATA guidelines. Therefore, therapeutic dissection of the lateral compartment is not controversial in patients with clinically proven metastasis, because this is likely to reduce the recurrence rate. However, there are still no guidelines for the optimal neck dissection range. Clinical outcomes and the complications associated with comprehensive neck dissection (generally level II to level V) have been reported in a number of studies, but the effects are still controversial.

This systematic review analyzed the distribution of LN metastasis in the most controversial components of comprehensive neck dissection, levels IIb and V. In addition, it is the first study to analyze complication rate, recurrence rate, and clinical outcomes by lateral neck dissection range. In this study, 13.7% of level IIb dissections and 22.1% of level V showed LN metastasis, which is similar to previously reported values [58]. Particularly in level V, Kim et al. reported a lower value of 8.6% when dissected without clinical evidence [26]. A particularly controversial region for neck dissection is level Va, the upper region of the lower border of cricoid cartilage. There were only three studies that analyzed the distribution of metastatic LN in level Va, and the metastatic LN distribution rate was reported to be 0–13%, which was lower than that of the overall level V distribution. As a result, levels IIb and V had comparatively lower metastatic LN distribution rates than other lateral compartments when there was no suspicious LN in that sublevel. Further research and analyses are needed through prospective large-scale studies that distinguish between levels Va and Vb.

After completing lateral neck dissection, the incidence of shoulder syndrome in the comprehensive neck dissection group from level II to level V was 1.5% to 27%. On the other hand, the incidence was 0% to 2.7% in the selective neck dissection group, which was extended from

Table 5
Clinical outcome according to lateral neck dissection range.

Author (Reference number)	Year	Study design	Methods to identify metastatic LN	Extent of LN dissection (n)		Follow up time (month)	Adjuvant treatment	Clinical outcome	Statistical comparison	Comments
McNamara et al. [42]	2016	Retrospective analysis	Macroscopic in operation, US, CT	Level II-Vb ± IIb, ND (364)	SND: III-IV ± II (120)	Median 63.5	RAI 278:86 (76.4%:71.7%) RT 8:1 (2.2%:0.8%)	Overall RFP 94.4%:89.4% Dissected neck field RFP 97.7%:89.4%	Overall RFP 0.158 Dissected neck field RFP $p < 0.01$	SND only in highly selective patient.
Kang et al. [28]	2014	Retrospective analysis	US(FNAB), CT	Level II-V ND (2 0 9)	NA	Median 54	RAI 209 (100%)	5-year OS 91.8% DSS 98.5% DFS 67.7%	NR	Isolated level IV involvement and no extranodal extension are candidates for SND.
O'Neill et al. [29]	2014	Retrospective analysis	US(FNAB),	Level II-V ND (1 2 1)	NA	Median 31	RAI 116 (96%)	DFS 66%	NR	
Zhang et al. [30]	2013	Retrospective cohort study	US(FNAB), CT	Level II-V ND (3 3 0)	NA	Median 108	NR	OS 95.8%	NR	Include level V in the therapeutic ND in high risk patients.
Turanli et al. [49]	2007	Retrospective analysis	US(FNAB),	MRND: I-V (32)	SND: II-IV (29)	Median 57.5:72	RAI 32:29 (100%:100%)	OS 58m:73m DFS 54m:56 m	OS $p = 0.33$ DFS $p = 0.92$	SND can be applied safely in selected cases.

LN, lymph node; US, ultrasonography; FNAB, fine needle aspiration biopsy; CT, computed tomography; ND, neck dissection; SND, selective neck dissection; RAI, radioactive iodine; RT, radiation therapy; RFP, recurrence-free probability; NA, not applicable; OS, overall survival; DSS, disease specific survival; DFS, disease free survival; NR, not reported; MRND, modified radical neck dissection.

level II to IV and occasionally to Vb. Although this pattern could not be tested statistically, this tendency can be expected because level IIb and level Va neck dissection require detection and exploration of the spinal accessory nerve. In a study by Kim et al., a retrospective cohort study reported that a significant number of shoulder syndromes were present in a comprehensive neck dissection group [26]. There were no other complications that showed differences according to the extent of neck dissection.

Only five studies have reported clinical outcomes according to the extent of lateral neck dissection. Because the results of these studies and follow-up period were variable, meta-analyses were impossible. McNamara et al. divided the studies into two groups: the comprehensive neck dissection group and the selective neck dissection group [42]. The comprehensive neck dissection group included levels II to V. Level II b was dissected if there were suspicious LNs in level IIa. In level V, only level Vb was dissected. The selective neck dissection group routinely included levels III and IV, but also included dissection of level II if clinically suspicious nodes were found at operation. In this study, although the dissected neck field recurrence-free probability significantly was lower in the comprehensive neck dissection group, overall recurrence-free probability did not show any significant difference. Turanli et al. also reported similar results with no differences in OS ($p = 0.33$) or DFS ($p = 0.92$) according to the extent of neck dissection. [49]. However, all studies were retrospective and the follow-up period varied, limiting our ability to draw conclusions.

As mentioned above, the effect of the extent of LN metastases on the lateral compartment on survival rate has been controversial, but it is known to affect the recurrence rate for WDTC. Therefore, in the 2015 ATA guidelines, LN metastasis of the lateral neck compartment was defined as a risk factor of structural disease recurrence. In cases of lateral compartment LN metastasis without extranodal extension, it was noted that the recurrence rate increased by 2%. In addition, the recurrence rate was reported to increase by more than 30% when the metastatic LN size was greater than 3 cm, and by more than 40% when extranodal extension was present [10]. Thus, literature reporting the recurrence rate according to the extent of LN dissection was identified and meta-analyses were performed. In the random-effects model, the recurrence rate was 11.2% in the comprehensive neck dissection group and 11.0% in the selective neck dissection group. McNamara et al. also reported that there were no differences in overall recurrence rate between the comprehensive neck dissection group and selective neck dissection group [42]. In that study, there was a significantly higher

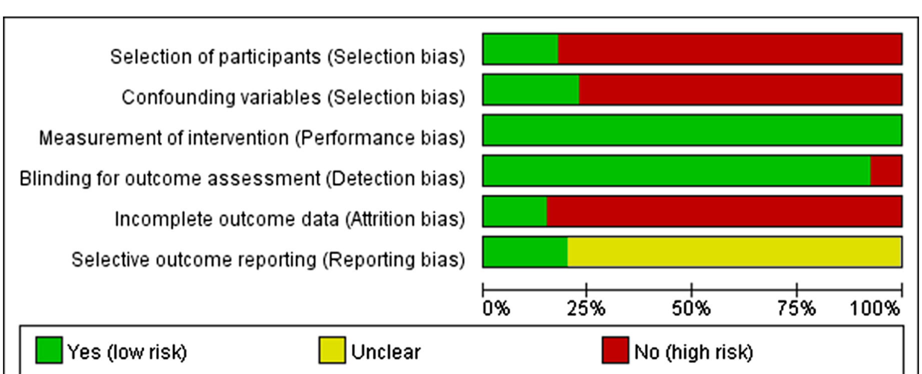


Fig 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

incidence of dissected neck recurrence in the selective neck dissection group, but the range of neck dissection was variable according to the clinical evidence, so it was excluded from the meta-analysis. Consequently, comprehensive neck dissection and selective neck dissection showed no significant difference in overall recurrence rate. However, in their study, adjuvant treatments such as radioactive iodine treatment were not consistently performed, limiting interpretations.

This systematic review and meta-analysis had several limitations. With the exception of a few studies, most were retrospective chart reviews, and therefore the level of evidence was low. In addition, there were only two comparative studies assessing clinical outcomes and recurrence rates according to the extent of neck dissection, and these were retrospective studies. Therefore, in several studies, fragmentation results were collected according to the extent of neck dissection, and the meta-analysis was only performed on the recurrence rate, thus limiting the interpretation of the results. In addition, the stage and other factors of WDTC are diverse and the number of operators varies, so bias should be considered. In the heterogeneity evaluation, the value of I^2 statistics showed a high heterogeneity of 84.944% to 86.042%. However, this systematic review and meta-analysis is meaningful in that it is the first study to analyze clinical outcomes, recurrence rates, and incidence of complications according to the extent of therapeutic lateral neck dissection. These results will be helpful in considering the extent of therapeutic lateral neck dissection.

Conclusion

Levels II b and V have relatively low metastatic LN distribution rates compared to other lateral compartments. The extent of neck dissection in lateral LN metastasis may not change the recurrence rate. Furthermore, comprehensive neck dissection and selective neck dissection were not significantly different in terms of overall clinical outcome. On the other hand, comprehensive neck dissection was more likely to cause shoulder syndrome than selective neck dissection. In conclusion, selective neck dissection should be considered in the limited cases in which there are no risk factors for tumor aspects or clinicopathological aspects that could affect prognosis.

Conflict of interest disclosure

None of the authors have a relevant financial relationship with a commercial interest.

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No involvement.

Appendix A. Supplementary data

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